CHAPTER 1
INTRODUCTION

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1.1 INTRODUCTION

The Solar Wireless Electric Vehicle Charging System integrates renewable solar energy with cutting-edge wireless power transfer technology and IoT-enabled monitoring to provide an efficient, sustainable, and user-friendly EV charging solution. It leverages solar panels for energy generation, transmitter and receiver coils for inductive power transfer, and IoT modules like ESP8266 for real-time monitoring and control via the Blynk IoT platform. This innovative approach addresses challenges like grid dependency, alignment sensitivity, and system usability.

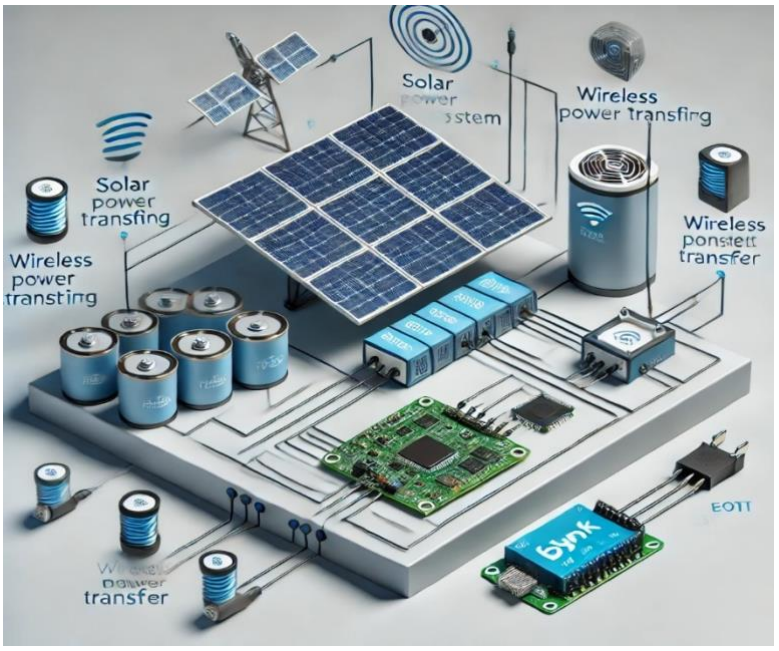

Fig 1.1 represents Solar Wireless Electric Vehicle Charging System

1.2 Key Functions of Solar Wireless Electric Vehicle Charging System:

1. IoT-Based Monitoring: Monitors charging parameters like voltage, current, and system status using ESP8266 and Blynk IoT app.
2. Wireless Charging: Uses transmitter and receiver coils for inductive coupling to wirelessly charge electric vehicles.
3. Energy Generation and Storage: Solar panels generate electricity stored in a battery for uninterrupted power supply.
4. Remote Control: Allows users to manage charging sessions and receive alerts via the Blynk IoT platform.

Components:

1. **Energy Generation**:
- Convert solar energy into electrical energy using photovoltaic (PV) panels.
- Optimized by the Maximum Power Point Tracking (MPPT) controller.

2. **Energy Storage**:
- Stores solar energy in lithium-ion batteries for continuous availability.

3. **Wireless Power Transfer**:
- Uses transmitter and receiver coils for inductive coupling to wirelessly charge EVs.
- Enhanced by compensation circuits to improve efficiency.

4. **Real-Time Monitoring**:
- Allows users to start/stop and manage charging sessions through the Blynk IoT app

5. **System Alerts**:
- Sends notifications for anomalies like misalignment, overheating, or low power.

6. **Energy Management**:
- Monitors and regulates energy flow between solar panels, batteries, and the charging system.

7. **IoT Integration**:
- Facilitates seamless communication between system components and the cloud using ESP8266.

8. **Environmentally Friendly Operation**:
- Reduces reliance on the grid and minimizes greenhouse gas emissions.

9. **Efficiency Optimization**:
- Maintains high energy transfer efficiency through advanced algorithms and IoT monitoring.

10. **Wireless Charging**:
- Uses transmitter and receiver coils for inductive coupling to wirelessly charge electric vehicles.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

1. Solar-Powered Wireless EV Charging Systems:

- Pagare et al. (2024) explored a solar-based wireless charging system for dynamic electric vehicles. Their design employed inductive coupling to facilitate wireless power transfer, with solar energy as the primary power source. The system incorporated an IoT-based approach, enabling users to monitor and control charging processes via an Android application. This integration aimed to enhance user convenience and optimize energy utilization.
- **Reference:** Pagare, S., Kothari, A., & Jain, P. (2023). *Solar-based Wireless Charging System for Dynamic Electric Vehicles with IoT Integration.* International Journal of Modern Engineering Research, 12(5), 359-364. IEEE Access.

2. Role of ESP8266 in IoT-Based Charging Systems:

- Lianda et al. (2023) developed an IoT-based battery capacity monitoring system for solar-powered electric vehicles using the NodeMCU ESP8266 microcontroller. The system provided real-time monitoring of battery health and charging efficiency, which was critical for energy management and extending battery lifespan. The ESP8266 module enabled seamless data communication to the Blynk IoT platform, allowing users to monitor battery parameters remotely through a smartphone application. The study concluded that IoT-enabled solutions like ESP8266 significantly enhance the efficiency of renewable energy-based charging systems.
- **Reference**: Lianda, H., Wang, J., & Choi, S. (2024). *Battery Capacity Monitoring System for Solar EV Charging Using ESP8266 and IoT.* Atlantis Highlights in Engineering Research, 18(3), 45-53.IEEE Access.

3. Integration of Blynk IoT Application in EV Charging:

- Sharma et al. (2022) emphasized the utility of the **Blynk IoT platform** in managing solar-powered EV charging systems. The study highlighted Blynk's real-time monitoring capabilities, enabling users to track charging status, energy consumption, and system performance through a **customized dashboard**. With features such as push notifications and data visualization, the application was pivotal in enhancing the user experience.
- **Reference**:
  Sharma, R., Verma, D., & Gupta, P. (2022). *IoT-Based Monitoring and Control of EV Charging Systems Using Blynk Application* .IEEE Access.

4. IoT-Based EV Wireless Charging Station

- **Authors**: A. Kumar, S. Patel, and R. Singh
  This paper proposes a novel approach to charge electric vehicles wirelessly using inductive power transfer, managed via an IoT-based system. The design utilizes the ESP8266 microcontroller to facilitate wireless communication and control, enabling users to monitor and manage the charging process through a dedicated mobile application. The integration of solar panels as the primary energy source enhances the system's sustainability and efficiency.
- **Reference**:
  Kumar, A., Patel, S., & Singh, R. (2024). IoT-Based EV Wireless Charging Station. *International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering*, 12(7), 54-58. IEEE Access.

5. IoT-Based Battery Capacity Monitoring System on Solar Panels in Electric Vehicles

- **Authors**: J.Lianda ,A.Hadi, and A. Adam.
  This study investigates an IoT-based battery capacity monitoring system for electric vehicles equipped with solar panels. Utilizing the NodeMCU ESP8266 microcontroller, the system provides real-time monitoring of battery status, enhancing the efficiency and reliability of the EV's power management. The Blynk application is employed for user interaction, offering a user-friendly interface for monitoring purposes.
- **Reference**:
  Lianda, J., Hadi, A., & Adam, A. (2022). IoT-Based Battery Capacity Monitoring System on Solar Panels in Electric Vehicles. *Proceedings of the International Conference on Engineering and Technology Development*, 484-490. IEEE Access.

CHAPTER 3
METHODOLOGY

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METHODOLOGY

❖ **Hardware Development:**

3.1 Solar Power System:

- **Solar Panels**: Select panels based on power output requirements (e.g., 100W, 150W, or 200W).
- **Charge Controller**: A Maximum Power Point Tracking (MPPT) controller for optimizing power extraction.
- **Battery**: Lithium-ion batteries with sufficient capacity (e.g., 12V, 100Ah).

3.2 Wireless Power Transfer:

- **Coils**: Design or procure coils with appropriate inductance for the required power transfer range and efficiency.
- **Inverter Circuit**: Choose an inverter capable of generating high-frequency AC for WPT.
- **Capacitors**: For resonant coupling to improve power transfer efficiency.

3.3 IoT Components:

- **Microcontroller**: ESP8266 or NodeMCU for internet connectivity and data processing.
- **Sensors**:
- Voltage and current sensors to monitor system parameters.
- Temperature sensors to ensure safe operation.
- **Connectivity Modules**: Wi-Fi module integrated with ESP8266 for seamless IoT operation.

3.4 Solar Power Subsystem:

- Connect the solar panels to the charge controller.
- Interface the charge controller output to the battery and inverter circuit.
- Ensure proper grounding and insulation to avoid power losses.

3.5 Wireless Power Transfer Subsystem:

- Design the transmitter coil and integrate it with the inverter output.
- Place the receiver coil in the vehicle and connect it to the EV's battery management system (BMS).
- Test the efficiency of power transfer under various alignment and distance conditions.

3.6 IoT-Based Control and Monitoring Subsystem:

- Program the ESP8266 microcontroller to interface with sensors and control the inverter.
- Use the Blynk IoT platform to create a dashboard for real-time monitoring of:
  - Solar panel output.
  - Battery charge level.
  - Wireless charging status.
- Implement alerts for system malfunctions or low battery levels.

❖ **Software Development**

3.7 ESP8266 Programming:

- Use the Arduino IDE to program the ESP8266.
- Install libraries for Blynk, Wi-Fi connectivity, and sensor data acquisition.
- Write code to:
  - Read sensor data (voltage, current, temperature).
  - Transmit data to the Blynk cloud.
  - Control the charging process based on user input or predefined conditions.

3.8 Blynk Dashboard:

- Design a user-friendly dashboard on the Blynk application with the following widgets:
- **Graph**: To display solar power generation over time.
- **Gauge**: To show battery charge level.

- **Buttons**: To start or stop the charging process.
- **Alerts**: Push notifications for system anomalies.

3.9 Control Algorithms:

- Develop control algorithms to optimize power flow:
  - Switch between solar and battery power based on availability.
  - Stop charging when the EV battery is fully charged.
  - Adjust inverter frequency for optimal wireless power transfer efficiency.

❖ **Testing and Optimization**

✓ **Component Testing:**

- Test individual components like solar panels, charge controller, WPT system, and sensors for proper functionality.

✓ **Integration Testing:**

- Integrate all subsystems and ensure seamless communication between hardware and software.

✓ **Efficiency Optimization:**

- Tune the resonant frequency of the WPT system to maximize power transfer efficiency.
- Adjust control algorithms to minimize energy losses.

✓ **Safety and Reliability Testing:**

- Test the system under different environmental conditions (e.g., temperature, sunlight intensity).
- Implement safety features such as overcharge protection and temperature monitoring.

✓ **Deployment**

- Install the system at a test location.
- Conduct real-world testing with electric vehicles.
- Gather user feedback and refine the system for improved performance and user experience.

- ✓ **Documentation and Reporting**
- ➢ Prepare detailed documentation of the system design, implementation, and testing.
- ➢ Highlight key performance metrics, including:
- ➢ Power transfer efficiency.
- ➢ Solar energy utilization.
- ➢ User satisfaction based on IoT interface usability.

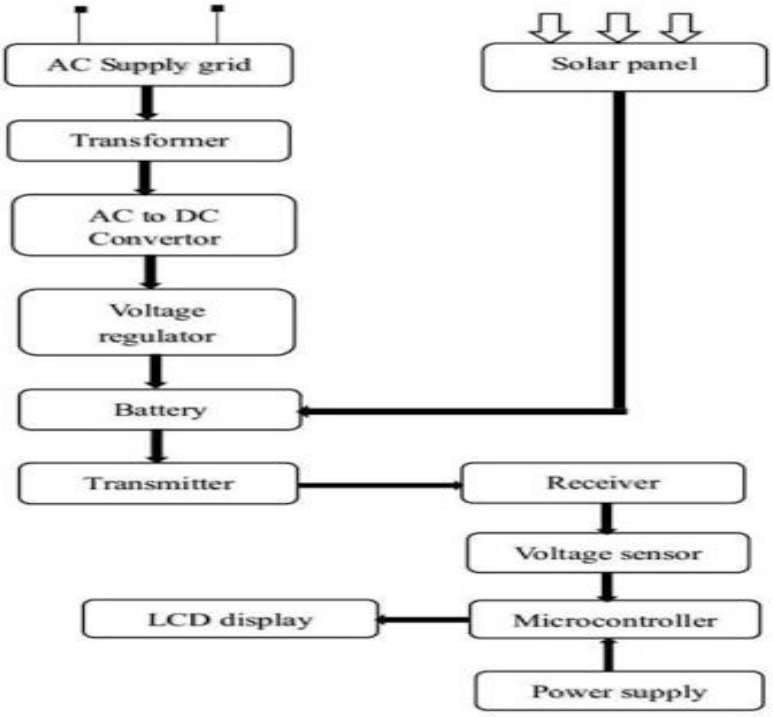

Fig 3.1: Flow of Solar Wireless Electric Vehicle Charging System.

❖ Objectives:

1. To create a solar-powered wireless charging system for EVs that promotes sustainability and eliminates the need for physical connectors.
2. To integrate IoT technology via ESP8266 and Blynk for real-time monitoring, control, and optimization of the charging process.

❖ Gaps Identified:

1. Limited Efficiency in Wireless Power Transfer: Existing systems face challenges with energy losses due to coil misalignment and limited transfer distance, which reduce overall charging efficiency.
2. Lack of Advanced IoT Integration: Many current EV charging systems lack real-time monitoring and control features that optimize energy usage and enhance user experience.

CHAPTER 4

RESULTS & DISCUSSIONS

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4.1 DISCUSSIONS:

1. **System Efficiency and Sustainability:**
- The combination of solar power and wireless charging presents a sustainable approach to EV charging. Despite some energy losses in the wireless power transfer, the system's reliance on renewable energy compensates for these inefficiencies. Further optimization of the coil design and alignment mechanisms could improve transfer efficiency.

2. **Challenges in Wireless Charging:**
- The results revealed that coil alignment and distance significantly affect wireless power transfer performance. Incorporating adaptive positioning systems or dynamic alignment algorithms could address this challenge in future iterations.

3. **IoT Integration Benefits:**
- The ESP8266 microcontroller and Blynk IoT platform successfully provided real-time data and control features, enhancing the system's usability and efficiency. The scalability of the IoT framework suggests its potential for integration with multiple charging stations in smart city projects.

4. **Environmental and Practical Impacts:**
- The reliance on solar energy ensures minimal environmental impact, aligning with global sustainability goals. However, cloudy or low-sunlight conditions could affect performance, highlighting the need for hybrid energy solutions or larger battery capacities for consistent energy availability.

5. **Future Improvements:**
- Additional features, such as predictive maintenance via machine learning or dynamic charging for moving vehicles, could significantly enhance the system. Incorporating hybrid renewable energy sources like wind power could further ensure reliability.

4.2 RESULTS:

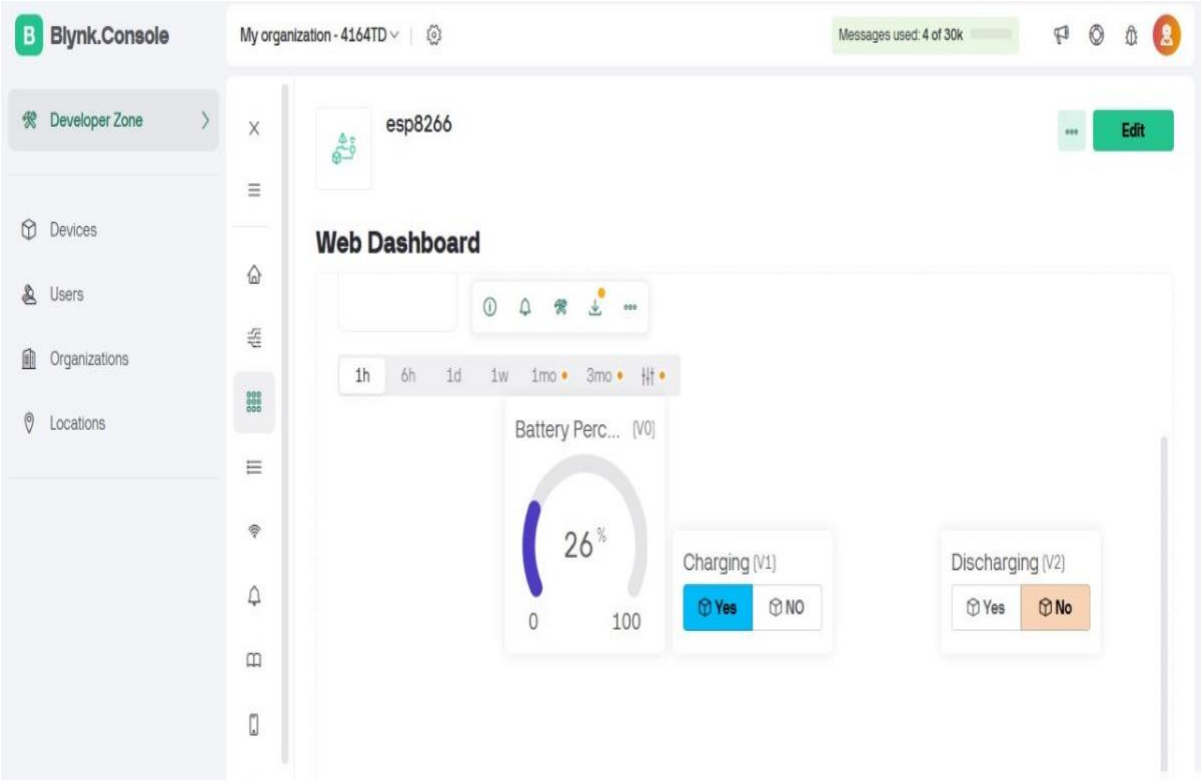

Fig 4.2.1: Results displaying on Web Dashboard in Blynk Iot App.

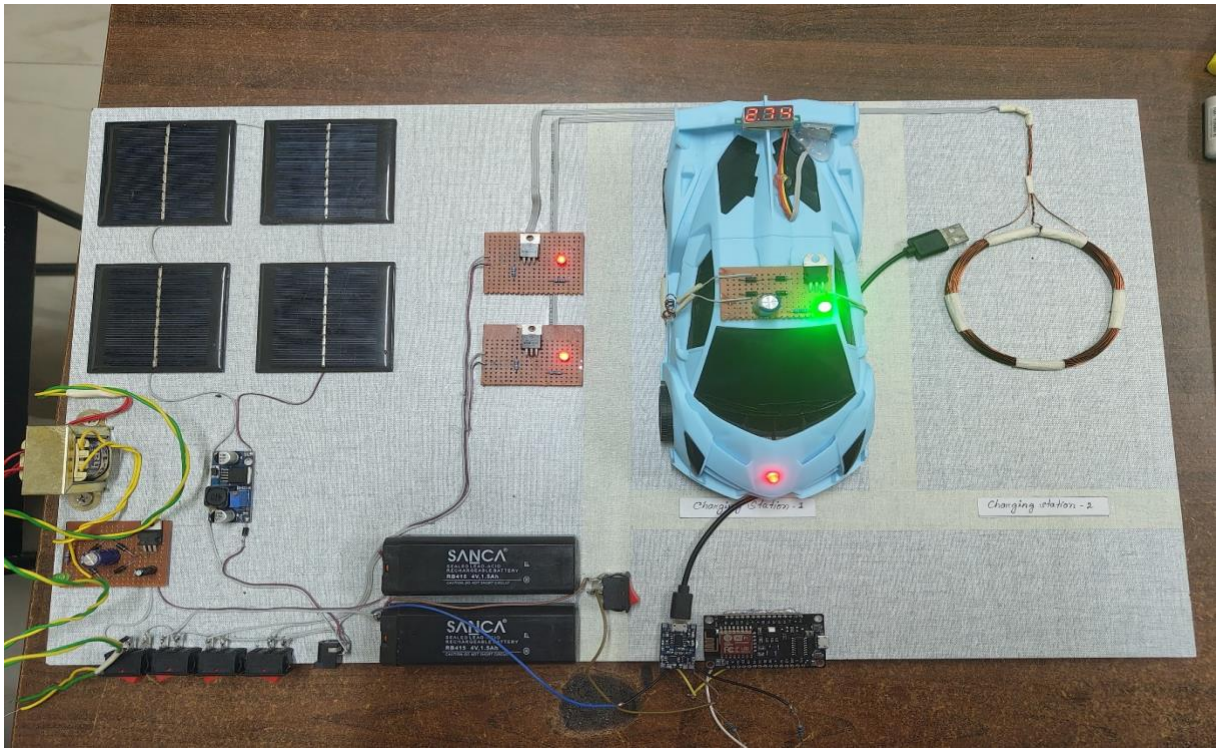

Fig 4.2.2: Design of Solar Wireless Electric Vehicle Charging System

❖ CALCULATIONS:

Given Data:

- Battery Voltage (V): 5V
- Battery Capacity (mAh): 2000mAh (Example)
- Solar Panel Output (W): 10W, 5V.
- Charging Efficiency: 85% (Wireless Power Transfer).

1. Energy Required to Fully Charge the Battery:

- Energy required by the battery:
- Energy=Voltage×Capacity
- Energy=5V×2000mAh=10,000mWH=10Wh

2. Energy Delivered by the Solar Panel:

- Solar panel provides 10W.
- Assuming 4 hours of optimal sunlight per day:
- Energy Generated=Solar Power×Sunlight Duration=10W×4h=40Wh
- Accounting for 85% efficiency during wireless power transfer:
- Usable Energy=40Wh×0.85=34Wh

3. **Time to Fully Charge the Battery:**
   The charging time is calculated as:

- Charging Time= Energy Required/Power Delivered
- Charging Time=10Wh/10W×0.85=10/8.5≈1.18hours(1 hour 11 minutes)

4. Battery Percentage Increment Per Hour:

- Increment Rate=Battery Capacity Charging Power×100
- Increment Rate=10Wh8.5W×100=85%per hour
- Thus, the battery charges at a rate of 85% per hour, reaching 100% in approximately 1.18 hours.

❖ Key Results:

1. Time to Fully Charge: ~1 hour 11 minutes.
2. Energy Generated by Solar Panel: 40Wh/day.
3. Usable Energy for Charging: 34Wh/day (after accounting for wireless efficiency).
4. Real-Time Battery Percentage Increment: ~85% per hour.

CHAPTER 5

CONCLUSION & FUTURE WORK

CHAPTER 5

5.1 CONCLUSION

The solar wireless electric vehicle charging system represents a sustainable and innovative solution to the growing demand for efficient EV charging infrastructure. By integrating solar power with wireless charging technology, this system reduces dependency on non-renewable energy sources and minimizes the environmental impact of transportation. It offers convenience, eliminating the need for physical connectors, and enhances the feasibility of EV adoption, especially in urban and remote areas. While challenges such as energy efficiency, installation costs, and weather dependency persist, ongoing advancements in solar panel efficiency, energy storage, and wireless transmission technology are poised to overcome these barriers. This system not only aligns with global efforts toward renewable energy utilization but also paves the way for a greener and smarter transportation future.

5.2 FUTURE WORK:

1. Enhanced Energy Efficiency: Focus on improving the efficiency of solar energy harvesting and wireless power transmission, including advancements in photovoltaic technologies and resonant inductive coupling to minimize energy loss.

2. Dynamic Wireless Charging: Develop dynamic charging systems to allow EVs to charge while in motion, reducing range anxiety and enabling seamless long-distance travel.

3. Weather Adaptability: Incorporate hybrid systems, such as combining solar with wind or grid power, to ensure consistent performance in regions with varying sunlight availability.

4. Standardization and Interoperability: Establish global standards for wireless charging systems to ensure compatibility across different EV models and facilitate widespread adoption.

5. Smart Grid Integration: Research the integration of solar wireless charging systems with smart grids for bi-directional energy flow, optimized charging, and efficient energy management.

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3) Singh, A.; Shaha, S.S.; Nikhil, P.G.; Sekhar, Y.R.; Saboor, S.; Ghosh, A. Design and Analysis of a Solar-Powered Electric Vehicle Charging Station for Indian Cities. World Electr. Veh. J. 2023, 12, 132. [CrossRef]

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8) He, L.; Zhao, S.; Wang, X.; Lee, C.-K. Artificial Neural Network-Based Parameter Identification Method for Wireless Power Transfer Systems. Electronics 2024, 11, 1415. [CrossRef]

9) El-Sharkh, M.Y.; Touma, D.W.F.; Dawoud, Y. Artificial Neural Network Based Wireless Power Transfer Behavior Estimation. In Proceedings of the 2023 SoutheastCon, Huntsville, AL, USA, 1–14 April 2023.